

Ideas Parked — Not Yet Merged (v11)

Staging document; each entry marks its eventual destination

This version reviews v10 against `PA.ReWrite.V15.22.tex` specifically, the version that actually carries the verb-inventory rewrite and the Section 2/Section 3 restructuring. Three kinds of change follow from that review, kept in separate sections below: terminology in v10’s own text that the rewrite has made stale (most of it symbols, one of it a genuine naming collision with a new formal operator); status markers on v8–v10 that said “not merged” and are now wrong, since the realignment they describe is the current paper; and gaps the rewrite itself opened by giving Collapse formal, axiom-level standing, none of which were trackable before that standing existed. Everything else in v10 was checked against the current paper directly and found unchanged and still accurate; it is carried forward below rather than merely cited, so this file stays a complete, standalone record.

1 Status update: the v8–v10 terminology work is merged, not scribbled

v8 proposed the realignment and flagged a symbol collision. v9 resolved that collision, provisionally, with $\ominus$ for UnCollapse. v10 revised that choice — $\ominus$ for UnCollapse, $\backslash$ for Remove — and called itself “ready to draft.” All three said, in their own closing status lines, that nothing had been written into any version of the paper yet.

That is no longer true. Checked directly: $\ominus$ appears zero times in `PA.ReWrite.V15.22.tex` — v9’s own proposed symbol was superseded by v10’s later choice, which is the one actually in the paper. $\ominus$ now means UnCollapse throughout; $\backslash$ means Remove throughout; $\oplus$ means Bundle throughout; Mix kept its own name after a brief detour through “MixMerge” and back. The full verb table v10 proposed is now Table 1 of the paper itself, with UnMix added to it by name. Every item v10 marked “Settled, this entry” — Collapse’s non-idempotence, the three convergent arguments for it, the N -storage account, $A^{(N)}$ notation, UnCollapse’s algebra at both boundary cases — is now the paper’s own stated content, not a scribble describing an intended future state.

[What this means for v8/v9/v10 below.] Their technical content was correct and is not repeated in full here; their status lines were right when written and are wrong now. Treat v8, v9, and v10 as a closed historical record of how the decision was reached, not as pending work. Kept, unedited, at the end of this file for that reason (Section 16) — their own words describe the reasoning accurately; only their framing as future work is stale, and this section’s note is the correction, not a rewrite of what they argued.

2 Stale symbols: v10’s own text, read against the current paper

The realignment changed what $\oplus$ and $\ominus$ mean, not just what they are called. Several of v10’s earlier sections — all written before the realignment was settled — use $\oplus$ for Gather’s old bowl-forming sense and $\ominus$ for Remove’s old sense, both now reassigned. Read literally against the current paper,

these lines say something false or confusing, not merely old-fashioned. Corrected readings, section by section:

- **The chladni citation (Section 6 below).** The quoted sentence — “Cross-mode coexistence is therefore $\oplus$ ” — is a direct quote from Paper 1, and the paper’s own words now read $\uplus$ at that exact sentence, confirmed directly against the current text. The quote below is corrected to match. Two more instances in that same section’s commentary (“already living on the $\oplus$ side,” “compositing unflattened layers is $\oplus$ ”) meant the same bowl-forming sense and are corrected the same way.
- **Meta vectors, link-vector grouping.** “Gathering, in the $\oplus$ sense” meant Bundle’s collecting act. Now reads: gathering, in the $\uplus$ sense.
- **The v5 identity-handle scribble, most seriously.** “Gather still satisfies $A \oplus A = A$ ” was true when written. Read against the current paper, where $\oplus$ is Collapse, this sentence is now *false* — Collapse’s whole point, settled in v10’s own later section, is that $A \oplus A \neq A$. The claim itself was never about Collapse; it was always about Bundle’s idempotence. Corrected: “Bundle still satisfies $A \uplus A = A$.” The same section’s “not collapsed by $\oplus$ ” and “under $A \oplus A = A$ ” need the identical fix, to $\uplus$, for the identical reason. One more symbol in the same section, “do not claim that $\{A\} \ominus A \dots$ inverts $\boxplus$,” used $\ominus$ for Remove, which is what the sentence is actually about; under the current symbol it reads $\{A\} \setminus A$. The underlying claims were correct throughout and remain correct; only the symbols were overtaken by a later decision these sections predate.

3 A genuine naming collision, not just a stale symbol

The v5 scribble’s own name for its mechanism — “uncollapse,” lowercase, informal — is a direct collision with UnCollapse, capitalized, now a formally defined operator with its own equation and algebra (Section 16, v10’s own entry). These are not the same thing, and the collision is not cosmetic:

v5’s mechanism: a pointer, written at the moment a bowl is read by Mix, from the resulting state back to the bowl that produced it. Looking the bowl back up later via that pointer. This is navigation by a kept record; it does not invert Mix, and v5 says so explicitly. It is a workaround for Mix’s own proven irreversibility, built entirely from meta-level bookkeeping (*id* fields, link vectors) that sits outside the 4-tuple.

The paper’s UnCollapse: removes one informational vector from a Collapsed atom’s own internal list, $N \rightarrow N - 1$, by direct computation on the atom’s own structure — no pointer, no bowl, no meta field, nothing external. It answers a different question (what contribution is taken back from a count-preserving combination) using different machinery (Axiom 7’s own vectors) for a different operator (Collapse, not Mix).

Why this matters going forward. v5’s mechanism is still a live, useful idea — a real answer to “can anything be done about Mix’s irreversibility,” and nothing above withdraws it. But it must never again be called “uncollapse.” Proposed name, used from here on: the **Mix-provenance link**. *id* stays the field name (unaffected — it was never a verb, so it does not collide); only the informal verb “uncollapse” retires, in favor of “recover via Mix-provenance” or similar. The tiny worked picture from v5 (bowl {red, green}, Mix reads yellow, provenance link from yellow’s *id* back to the bowl’s) is unaffected in substance and is restated below under the corrected name.

4 Three gaps the rewrite opened, not yet tracked anywhere

None of these existed as trackable items before Collapse had formal, axiom-level standing. All three are visible directly in the current paper’s own text — two as explicit, stated omissions; one as a silence worth naming rather than a stated one.

Collapse across differing sorts (the paper already says this is open). Quoted directly: “Axiom 7 currently requires every vector in one atom to share the atom’s single τ ... Admitting a per-vector τ_i would be the natural extension ... but it is not adopted here and is left open.” This is Table 2’s own open[¶] entries for every cross-sort pair. Worth tracking here specifically because it is the same shape of question as Gap 1 (Section 6) for a different operator, and the two should not be assumed to share an answer: Mix blends, so refusing across directions protects against manufacturing a direction that was never there; Collapse preserves every contribution without blending anything, so the case for refusing it across sorts is not the same case, and has not been separately made.

Collapse as T ’s actor: not claimed, not ruled out. Quoted directly: “every worked T path in this paper takes one [teaching atom], and a Collapsed-atom-as-actor example is not claimed here.” T ’s machinery for accepting a richer actor already exists in principle — V_{derived} , or beyond three dimensions, $N_{\text{dim}} - 2$ of the Collapsed atom’s own orthonormal arrows directly — but no example walks a Collapsed atom, and the cast that would need naming at the call site is not worked out anywhere.

Assembly and Collapse: no stated relationship at all. Checked directly: Axiom 6 (Assembly) does not mention Collapse once, in either direction. Whether a Collapsed atom may stand as one of Assembly’s two related units, whether U_{AB} means anything when A or B carries $N > 1$ internal vectors, is not addressed, not refused, not flagged as open — simply silent. Of the three gaps here, this is the one with no sentence in the paper pointing at it; recorded so it is not mistaken later for a settled “no.”

5 A new route worth recording, not a solution

The “reducing a path by merging close actors” open item (Section 9 below, unchanged and still open in the current paper) asks, word for word, whether a representative actor should be “a $\boxplus$ -blend of the close actors, an average rotation, or something else.” Collapse did not exist as a named option when that sentence was written. It is a structurally plausible candidate for “something else,” worth recording precisely because of how it would differ from the other two: Collapse the close actors losslessly first — every contribution kept, nothing blended, nothing decided yet about direction — and only then define a second, reading operation (the way Mix already reads a Bundle) that would extract a single representative direction from the Collapsed result.

This is not a solution. The reading step that would turn a Collapsed atom into one representative direction does not exist, is not sketched here, and may turn out to face the same direction-blending objection that keeps Gap 1 open — extracting one direction from several is exactly the operation Route 1’s chladni evidence counsels refusing. What Collapse changes is only that the open item’s own “or something else” now names something specific, where before it named nothing. The same observation extends, with the same limit, to the v6 item (Section 14 below): for same-sort regions it offers an alternative to the already-working Hill bijection, not a need; for cross-sort regions it

offers nothing new, since Collapse cannot yet cross sorts either (the first gap above) — the hard half of v6 stays exactly as open as Gap 1 itself.

6 Destined for Pattern Arithmetic (still formally open — unchanged, confirmed against the current paper)

Mix across differing directions. Checked directly: the paragraph’s own text in `PA_ReWrite.V15_22.tex` still reads “not settled here,” still lists the same three unchosen routes, still carries the same with-drawn provisional recipe. Nothing here changed it, on explicit instruction to leave gap work alone until terminology settled; recorded accurate and current, not merely inherited.

[Found: the paper already answers this question once, for one sort, and points toward a general answer.] The chladni inclusion states, word for word (corrected symbol, per Section 3’s sibling correction above): “Across modes the map is not available, and not for want of effort. Mode space is not closed under addition... Cross-mode coexistence is therefore $\uplus$ — a bowl of modes — and the resulting figure is a reading of that bowl which is not $\boxplus$. This is the open question of [the differing-directions open item] met in a concrete domain, and it suggests the question may be closed negatively for this sort rather than left waiting.” This is a citation, not a new argument — the paper’s own words, in a section that was not previously read as bearing on Gap 1 directly.

[Why this converges with everything else worked through in this project, not just chladni.] Once Mix is read as “resolve several reports on one shared question into a verdict” (argued in v2/v3, carried forward unchanged), refusing across-direction combination stops looking like a gap and starts looking like the correct behavior of an operator that was never asked that question in the first place. And every practical workaround that actually worked in this project was already living on the $\uplus$ side of precisely this line, independently: bowls holding separate weather channels, per-channel similarity scoring, the candlestick summary for reference-atom building, Assembly’s own camp-to-system Hill bijection. None of them ever needed Mix to cross a direction. All four arrived, separately, at “keep it in a bowl, read it there” — which the chladni section had already stated as a formal result.

[Status: a leading candidate, not yet a decision.] This supports Route 1 (refuse outright, from the three already on file) over Route 2 (rotate first) or Route 3 (two-stage operator) as the likely resolution, with a real citation behind it rather than preference. Writing this into Paper 1 would mean editing the “Open: Mix across differing directions” paragraph to state the refusal explicitly, citing the chladni section as precedent. Not done here, on request — parked, ready, pending a decision to merge.

[Carried forward unchanged: the reading argued in v2/v3.] Mix is not “multiple vectors becoming one vector.” It takes several reports already agreed to answer one shared question and returns a verdict about that question. The shared $\hat{n}$ is the precondition that makes “these answer the same thing” true, not a boundary condition on a more general operator. Supporting evidence: same- direction opposite-phase cancellation to black (principled contradiction-handling, not a fabricated compromise); unmixing’s proven ill-posedness, with metamerism as independent real-world precedent; the paper’s own opening line, “two notes yield a chord, not a larger integer... the word ‘add’ names a different rule.”

[Carried forward unchanged: the “we mix all the time” challenge, resolved.] Sugar in tea is one dial, never needed differing directions. Red plus green additively sits “at the same place” in the paper’s own words — already inside Mix’s settled domain. Paint-layer flattening is confirmed, across every source checked, as destructive and irreversible, matching unmixing exactly;

compositing unflattened layers is $\uplus$, not $\boxplus$, which is why “unmerge them later” was always possible. Representing a whole painting as one vector remains Gap 1 itself, not a counterexample to it.

[Carried forward unchanged: the two-part split, now read as one part likely answered.] Embedding validity — does geometric closeness on a shared sphere track real semantic agreement at all — was the harder, prior question, true for hue by deliberate construction, unstated for an arbitrary sphere of concepts. Given the chladni precedent above, the likely answer for sorts where this was never stated is that no valid embedding needs constructing, because cross-direction combination is refused before the embedding question would even matter.

[Carried forward unchanged: still confirmed not to help.] Magnitude self-bounding near opposition does not license the direction that comes with it. The Hill form and Assembly’s Hill bijection both fix magnitude only, verified twice, never direction.

7 A correction, recorded so it is not repeated

τ_{weather} names two different problems, and only one of them is Gap 1. Raised once as a proposed admissible sort for combining different weather concepts (rain, drought) across directions — that question is genuinely Gap 1, and the discussion above bears on it. Raised a second time, later, meaning something narrower: given several already closely-aligned vectors within one bounded region, how to arrive at a single representative atom, using the Hill form. That second question does not invoke $\boxplus$ across differing directions at all — it is Assembly’s already-verified system-promotion mechanism (16 camps to 13 systems under the Hill bijection), recognized in a new setting, not a Paper 1 admissibility question. The two were briefly conflated in this project; recorded here so the same conflation is not repeated. Assembly’s version needs nothing further; Gap 1’s version is addressed in Section 6 above. Unaffected by the rewrite; checked, no terminology in this section needed any update.

8 Destined for Pattern Arithmetic (advances an existing open item, unchanged)

Location as a sort — a third candidate. The existing paragraph names two candidates and adopts neither. Confirmed present and unchanged in the current paper: `PA_ReWrite.V15_22.tex`’s own “Open: location as a sort” paragraph still asks the identical question in identical terms. A third was tested here: dedicate separate, disjoint dimensions to content, to time, and to location, rather than letting any of them share an axis.

[Verified.] Two shifts acting through disjoint content dimensions commute exactly (checked numerically, not approximately: time-then-location and location-then-time landed on the identical point to full floating-point precision). Two shifts forced to share a single content dimension do not commute, confirming the disjointness is what matters, not rotation itself.

[Verified.] Content stays recoverable after both shifts: projecting back onto the content dimensions alone reproduced the original direction closely, while the time-axis and location-axis each carried a clean, separately-readable amount of shift, with nothing blended between them.

[Open, unchanged.] The encoding itself — what specific rotation a given time value or location corresponds to — is not stated anywhere. The geometry now supports it cleanly; the mapping does not exist yet.

This is real, checkable content, not yet written as an addition to the existing paragraph.

9 Destined for Pattern Arithmetic (open item stays open, unchanged, now with a newly available route noted above)

Reducing a path by merging close actors. The existing paragraph asks for three things: a stated distance, a merge rule, and a bound on how far a reduced path’s endpoint may drift from the unreduced one. Discussion here supplied a distance (see below) and partially addressed the merge rule, but only for magnitude, not for direction — and direction is the harder half this item is actually about. Confirmed present, unchanged, in the current paper; see Section 5 above for what its existence newly suggests, not yet a resolution.

[**Verified.**] Switching the strength calculation from the companion paper’s clip to Inference’s already-established Hill form, $r' = |z|/(K + |z|)$, changes nothing about the resulting direction $\chi' = \arg(z)$. Checked directly on two constructed sequences with identical content and opposite order: both gave the same angle under either formula. The Hill substitution fixes evidence-count blindness in strength; it does not touch the order-blindness this open item is actually asking about.

[**New: a candidate grounding for the stated distance, item (1) of the three things required.**] How many actors can be mutually distinguishable — separated by at least some angle $\theta_{\min}$ — on $\mathcal{S}$ at all is a real, named problem (spherical codes, the Tammes problem), not an arbitrary choice to make. Exact optimal counts are known only for small N ; a standard approximate bound, treating each point as owning a non-overlapping cap of half-angle $\theta_{\min}/2$, gives

$$N_{\max} \approx \frac{4\pi}{2\pi(1 - \cos(\theta_{\min}/2))} = \frac{2}{1 - \cos(\theta_{\min}/2)}.$$

Computed for illustration:

$\theta_{\min}$	approx. max. distinct points
1	52,525
5	2,101
10	526
30	59
60	15
90	7
120	4

This is an approximation, not an exact extremal result, but it is a real, checkable mathematical object rather than a guess: naming a $\theta_{\min}$ for “close enough to merge” directly bounds how many genuinely distinct actors can exist near each other at all, which bears on item (3), the drift bound, as well as item (1).

[**Open, unchanged.**] No merge rule for direction exists. No drift bound exists for either magnitude or direction. This item remains exactly as open as it was before this discussion, now with the magnitude half more precisely separated from the direction half, a real candidate supplied for the distance half specifically, and one newly nameable candidate mechanism for the merge rule itself (Section 5 above), not yet worked out.

10 Destined for Inference (new mechanism, unchanged)

Candlestick summary: a richer merge rule for reference-atom building. Not a resolution of the Paper 1 item above — a different question. Paper 1’s open item is about *walking*: can a shortened path still be walked to roughly the same endpoint. This is about *reading*: how to

summarize a group of atoms for comparison against a reference atom, which is Inference’s own established business (its reference-atom section), not T ’s. Destined for a different paper; unaffected by anything in this review.

[**Verified.**] Open/close/high/low, computed the way a trading candle is, distinguishes exactly the case flat Mix could not. Two sequences, identical content, opposite order — naive Mix gave the identical angle (47.5) for both. The candle’s open and close swapped between them ($20 \rightarrow 160$ versus $160 \rightarrow 20$), correctly reading one as ending badly and the other as ending well. High and low stayed identical for both, correctly capturing that only the *range* was shared, not the *direction of travel*.

[**Settled, matches existing machinery.**] The scope rule — every atom belongs to exactly one time unit, no overlap between buckets — is not new; it is precisely the discrete-time-bucket metadata already defined on a walk’s own steps in the companion paper.

[**Open.**] This gives Inference a genuinely better reference-atom merge rule than flat Mix. It is not proposed as a fix for Paper 1’s walking-specific open item, which remains untouched above. Whether open/close/high/low itself needs a coherence check analogous to $\bar{R}_c$ has not been considered.

11 Destined for Inference (naming caution, unchanged)

“Inference Mandelbrot” is not Assembly’s Mandelbrot, and should not share its name. Assembly’s multi-zoom promotes an unordered thing (a settled camp) and is fully handled by the Hill bijection already verified there. Whatever would promote an *ordered* thing (a walked path, a movie, a time series) into one higher-level atom is a different, harder problem — it is the direction half of the path-reduction item above, restated from a different angle. [**Naming flag only.**] If this is ever built, it wants its own name; reusing “Mandelbrot” for it would be the same kind of collision already caught and corrected for “collapse,” “singularity,” and the quantum swap test — and now, per Section 3 above, for “uncollapse” as well.

12 Destined across three papers (new, split by piece; symbols corrected, substance unchanged)

Meta vectors: a second kind of vector alongside information vectors. Proposed: an information atom may carry two distinct kinds of vector. Information vectors are Axiom 7’s bundle v_i ’s, capped at $r_i \leq 1$, potentially several sorts τ across a whole record. Meta vectors describe the atom rather than being part of its content — three named so far: V_{derived} , link vectors (pointers between atoms, one- or two-directional), and a timestamp vector. Explicitly left open whether more meta-vector types are needed. Meta vectors are not capped at 1, and grouping several atoms’ meta vectors together on like terms is proposed as a distinct mechanism for building higher structures. This is one idea that lands in three different places once traced to where each piece actually belongs.

[**Destined for Pattern Arithmetic.**] The taxonomy itself — information vectors versus meta vectors as a formal, named distinction sitting on top of Axiom 7 — is structural, the same kind of addition Axiom 7 itself was. V_{derived} already lives there. A timestamp vector, if adopted, would be a third, genuinely different treatment of time from the two already on file above: distinct from the discrete-bucket metadata already defined on a walk’s own step, and distinct from the dedicated-axis encoding in the location-as-a-sort entry. Not yet clear whether more than one of the three should coexist, or which.

[**Destined for Assembly.**] Link vectors specifically. This is the mechanism the shelf’s own neighbour-graph note was already naming without a concrete form: “a data structure on seats after rest, not T .” Grouping link vectors on like terms is collecting edges into a graph, not merging them — gathering, in the $\uplus$ sense — so it carries none of the open questions below it.

[**Destined for Inference.**] The grouping mechanism itself: querying several atoms’ meta vectors, confirming they sit close together, and promoting one shared meta-vector up to the group as a whole. This is the space-time gating idea from earlier discussion, generalized into a standing principle. Timestamp and location get *used* here; they are only *defined* in Pattern Arithmetic above.

[**Verified, carried forward as a caveat rather than a new finding.**] Combining several atoms’ V_{derived} ’s stays safe only under the constraint the original already carries: auxiliary, never promoted to a trusted, walkable atom on its own. Checked directly that neither Inference’s Hill form nor Assembly’s Hill bijection touches this constraint — both fix magnitude only, never direction — so grouping V_{derived} ’s inherits that same limit rather than escaping it.

[**Renamed here, per Section 3 above.**] Link vectors were separately discussed as one possible mechanism for a limited, honest form of Mix-provenance recovery: not inverting Mix (still impossible, unaffected by anything here), but keeping an explicit, external pointer from a Mixed result back to the bowl that produced it, created at Mix-reading time, before the bowl itself might otherwise be discarded. This is navigation via a kept record, not reconstruction from the Mixed result — the distinction the white-light/prism discussion turned on, confirmed there to matter. No longer called “uncollapse”; see Section 3 for why.

13 Destined across three papers (scribble, v5; renamed and symbol-corrected, substance unchanged)

Identity handle as a meta field, for Mix-provenance recovery. An information atom may carry one more meta quantity: an opaque identity id , assigned before any Bundle or Mix, and kept on the atom afterwards. The scribble that prompted this note called it a “meta vector” with τ_{id} equal to that identity. Fine-tuning below refuses τ_{id} as a sort and keeps id as a *field*, in the same shelf as timestamp, mint-location, and link pointers already named in the meta-vector entry.

[**What it is for.**] Mix remains irreversible (Paper 1 v1.1). Mix-provenance recovery here does not reconstruct ingredients from a yellow. It means: at Mix-reading time, write a pointer from the resulting atom (or from the register occupant after a walk) back to the bowl that produced it, and keep that bowl under the same id (or a pair $result_id \leftrightarrow bowl_id$). Later, given the result, look the bowl up. Navigation by a kept record, not inversion. This is the same distinction already recorded under link vectors and the white-light / prism discussion.

[**Viability — what works.**] An opaque handle is the right shape. It does not need to live on $\mathcal{S}$, does not need $r \leq 1$, and does not need a direction. A string or integer key is enough. Creating the pointer *before* the bowl is discarded is the whole trick; after discard there is nothing to recover. Links already named that graph; id is the node name those edges need. Assembly can own the edge table (neighbour / parent-bowl). Paper 1 only needs the taxonomy: id is meta, not a unit of truth. Inference may use id to re-open a bowl and run a second reading.

[**Viability — what to refuse now.**] Do not mint τ_{id} . τ is the legality tag of an information vector. An identity is not a pattern and must not appear in the type table. Do not encode id as an angle on the sphere; colliding two ids geometrically would be a false Mix. Do not claim that $\{A\} \setminus A$ or T^{-1} plus id inverts $\boxplus$. Bundle still satisfies $A \uplus A = A$: two tokens of the same pin share one address and, if they are the same *instance*, one id ; two distinct instances of the same pin

(two physical chairs) need two ids and are not Bundled by $\uplus$ unless the lexicon says they are the same state. That instance-versus-kind cut is the fine-tuning this scribble does not finish.

[**Tiny picture.**] Bowl $B = \{\text{red}, \text{green}\}$, $\text{id}_B = \#17$. Mix reads yellow, $\text{id}_Y = \#18$, meta link $\#18 \rightarrow \#17$. Looking up $\#18$'s provenance returns B , still red and green, still two atoms. The yellow is not pulled apart. The prism was kept on the shelf.

[**Status.**] Scribble. Destined: taxonomy to Paper 1 (meta shelf, next to V_{derived}); the pointer table to Assembly; lookup-and-reread to Inference. Not merged. Fine-tune later: who assigns ids, instance vs kind under $A \uplus A = A$, and whether one atom may keep a stack of parent-bowl ids after several Mixes.

14 A theoretical problem, split the same way an earlier section already split one — v6, unchanged except for the new route noted above

Can every atom in a unit space, across all labels and types, be merged to one representative atom? Posed precisely: a bounded region holds many already-settled atoms, of possibly *differing* sorts τ , not just differing content within one sort. Can the whole collection become one representative for that region? Framed, as requested, with both a theoretical and a practical answer in view, the way the sphere-packing question above was.

[**Solved — but only for the restricted case, which is an earlier correction's general form.**] When every atom in the region shares one sort (or, more precisely, one already-admitted φ_τ direction-family), this is exactly Assembly's already-verified system-promotion mechanism: many close atoms, one representative, via the Hill bijection rather than the naive clip, because Hill preserves evidence-count where the clip saturates and loses it. Sixteen camps correctly became thirteen systems on this basis, not eleven. "A lot of random atoms" changes nothing here — randomness in position or content is fine, provided the sort is shared. This is the τ_{weather} correction, restated as the general pattern it always was: random instances of one τ , closely aligned, merge cleanly. Section 5 above notes Collapse as an alternative route for this same-sort case specifically — not a need, since Hill already works here, but a newly available option.

[**Still open — this is Gap 1, not a second result.**] "Across all labels and types," taken literally, names the harder case: atoms of genuinely *different* sorts in the same region. This is not solved by Hill's equation or the bijection, and — confirmed by Section 4 above — not solved by Collapse either, since Collapse cannot yet cross sorts for exactly the same reason Mix cannot. Confirmed twice already in this project: both Inference's Hill form and Assembly's Hill bijection fix magnitude saturation only, and neither touches direction or sort. Merging across genuinely different sorts into one atom is precisely Gap 1 (Section 6), for which a leading candidate (refuse outright, per the `chladni` precedent) exists but has not been written into Paper 1.

[**Theoretical versus practical, for the cross-type half specifically.**] The sphere-packing question above had a well-defined continuous quantity (point count) that a practical approximation could meaningfully estimate. This question's open half does not: there is no stated φ_τ for combining across sorts to approximate in the first place. A random or approximate merge rule cannot stand in for a merge rule that has not been defined — this is an admissibility gap, not a precision gap, and the two should not be read as the same kind of "open" question.

[**So, precisely, correcting the framing in which this was posed.**] "We solved it using bijection and Hill's equation" is true for same-sort regions, and is a real, verified, reusable result — but it is not yet true for atoms spanning genuinely different types, which remains exactly Gap 1, unresolved by anything checked so far. The two cases should be kept as separate claims, the same

way an earlier section already separated them once for τ_{weather} specifically; this entry generalizes that same correction rather than adding a new one.

15 Destined for Pattern Arithmetic (scribble, v7; unchanged, not yet merged)

“Refuse outright” is a claim about $\boxplus$, not a permanent bar on ever combining unlike sorts by any mechanism. Raised as a caution on Gap 1’s leading candidate: likely that only like-typed atoms will ever practically need merging, but unwise to state Route 1 in a way that forecloses future, explicitly-justified mechanisms for binding unlike sorts, should a practical need for one arise. Checked against the current paper: still not written into the “Open: Mix across differing directions” paragraph; still exactly as scoped a caution as when drafted.

[The distinction that matters.] Route 1 was always a claim about $\boxplus$ specifically — that this particular operator, as defined, was never asked the cross-direction question and correctly declines to answer it. It was never a claim that no mechanism, under any name, could ever combine unlike sorts. Conflating the two would overclaim something the evidence never supported.

[Why this is not a new principle, but a confirmation of one already load-bearing in the paper.] Axiom 0 is already built this way: “a sort absent from this table is undecided, not admitted; treat it as inadmissible until a φ_τ is stated.” Inadmissible *now*, for lack of a stated proof — not inadmissible forever, by declaration. `chladni` is the existing demonstration: admitted later, on stated grounds, not pre-emptively barred. Even `chladni`’s own hedge on cross-mode combination, the strongest evidence behind Route 1, uses “may be closed negatively... rather than left waiting” — a leaning, not a verdict.

[How Route 1 should actually be worded, if merged.] Not: “ $\boxplus$ refuses states at differing directions, full stop.” Better: “ $\boxplus$, as currently defined, refuses states at differing directions; a cross-sort reading remains possible in principle, as its own stated mechanism, admitted on its own terms the way any new capability is — not as a modification of $\boxplus$ itself.” This keeps the refusal honest about what has actually been shown without overclaiming permanence the evidence does not support.

[Status.] Scribble. A scoping note on Gap 1’s existing entry, not a new result and not yet written into that section’s own text. “Probably rare” (practical likelihood) and “permanently barred” (structural impossibility) are different claims; only the first has support, and Route 1’s eventual wording should reflect only that one.

16 Closed history: v8, v9, v10, kept verbatim

The three sections below are reproduced unedited from v10. Their technical content is correct and is now the paper’s own content, not a proposal; their own status lines (“scribble,” “not merged,” “awaiting a decision”) describe their state at the time of writing, not the state today. See Section 1 above for the correction. Kept here, word for word, as the historical record of how the current verb inventory was actually reached — useful for exactly that reason, and for no other: nothing below should be read as open work.

v8 — proposed terminology realignment, with cost assessment (historical)

Proposed: Collapse, UnCollapse, Bundle, MixMerge, UnMix. A full realignment of five terms, proposed together:

- **Collapse** ($\oplus$) — one atom, one origin, several internal vectors.
- **UnCollapse** ($\ominus$) — subtract one atom’s vectors from another; not previously discussed in depth.
- **Bundle** — a bowl/set, many atoms, kept separate, separate origins.
- **MixMerge** ($\boxplus$) — one new atom, fused.
- **UnMix** — not possible.

This reassigns Collapse and Bundle relative to the reading settled two exchanges earlier in this project (Collapse = $\oplus$ = the bowl, confirmed three times directly from source text and from the bowl’s own definition as a separate finite set). The proposal here reverses that assignment. Recorded as a deliberate choice to evaluate, not adopted, and not written into Paper 1.

[Blocking: a direct symbol collision.] $\ominus$ is not available for UnCollapse. It already carries its own formal axiom: “Axiom 4: Gather and remove ($\oplus$, $\ominus$)... $\ominus$ undoes a gather. It is not anti-phase interference.” Two incompatible operators cannot share one symbol in the same document. A different symbol is needed before UnCollapse can be merged under any sequencing; this is not a matter of care in editing, the symbol itself is already spoken for.

[Instance counts, checked directly against the source.]

Term	Current meaning	Count	Title usage
Collapse	$\oplus$ ’s bowl-forming act	14	Yes — a subsection title
Mix / $\boxplus$	fused, single output	77 / 50	Yes — Axiom 5’s own title
Bundle	one atom, N internal vectors	10	No
Bowl	a set of separate atoms	80	No, used throughout

[Cost, per piece.]

- *Collapse as one atom*: high cost, and a reversal rather than a rename. The “Process principle: Collapse, path, result” subsection’s own stated reasoning — “collapse is the act that discards the order and count of that sequence... R = the bowl” — would need rewriting, not relabeling: its current logic explains why collapse yields a set, and that explanation would need to become its own opposite.
- *Bundle as bowl*: high cost. Collides with Bundle’s own existing, different meaning (Axiom 7’s N -vectors-in-one-atom). Bowl already carries 80 instances of its own, separate meaning; this is a reconciliation of two already-distinct, already-used terms, not an added synonym.
- *MixMerge, rename only, same underlying behavior*: moderate cost, mechanical. 77 instances is substantial, but nothing about $\boxplus$ ’s actual behavior changes — risk is missed instances or inconsistent phrasing, not logical breakage.
- *UnMix, named but still impossible*: low cost. The concept is already stated (“division as unmixing is shown ill-posed”) — this mostly formalizes an existing result with a name and a symbol, once a free symbol is chosen.

[Overall, honest verdict.] The $\ominus$ collision is a hard blocker, independent of sequencing, and needs resolving before anything else proceeds. Past that, Collapse-reversal and Bundle-reassignment are the expensive, high-risk pair, each touching foundational, load-bearing text and interacting with one another. MixMerge and UnMix are comparatively cheap and could move first, independently, as a lower-risk starting point if this is pursued.

v9 — symbol resolution for Bundle and UnCollapse, plus when ids are actually needed (historical)

Bundle gets a symbol and becomes a verb.

$$\text{Bundle : } A_1 \uplus A_2 \uplus \cdots \uplus A_n \longrightarrow B \quad (\text{many atoms, kept separate}).$$

Checked clean against the source: $\uplus$ (multiset union) does not appear anywhere in Paper 1. The choice is principled, not arbitrary — $\uplus$ ’s standard mathematical meaning, combine while preserving separate membership, is exactly Bundle’s job. Worth being precise about what this move actually is: with $\oplus$ reassigned to Collapse under the v8 scheme, Bundle-as-verb takes over the role Paper 1’s own $\oplus$ currently performs (the bowl-forming act, nothing merged, nothing lost). Not a new mechanism — a new name and symbol for an existing one, freed up because the old symbol moved.

UnCollapse gets a symbol. $\ominus$, also checked clean against the source. Visually related to $\ominus$, so the family resemblance between Collapse/UnCollapse reads correctly, while remaining a genuinely distinct symbol — no collision with Axiom 4’s own $\ominus$. *[Superseded by v10 below: the symbol actually adopted is $\ominus$ itself, not $\ominus$, once Remove moved to $\backslash$.]*

[Precisely when ids are needed for UnCollapse, and when they are not.] Collapse, under this scheme, is lossless by construction: every v_i stays individually intact inside the resulting atom. Splitting it back apart is *already* structurally possible without ids whenever every incoming vector has a distinct $(r_i, \hat{n}_i, \chi_i, \tau_i)$ — the bundle’s own listing already carries everything needed.

Ids earn their place in exactly one case: two *distinct* source atoms that happen to share identical values. Without an id, the two resulting entries inside the bundle are indistinguishable from each other — UnCollapse would return the right values, with no way to confirm whether this was genuinely two separate instances that happened to match, or something else that produced the same numbers. This is the same instance-versus-kind distinction already on file from the v5 scribble: “two distinct instances of the same pin... need two ids.”

So, stated precisely rather than as a blanket claim: UnCollapse is value-exact with or without ids. What ids add is identity-exact recovery — confirmation that what comes back out is traceably the same atom that went in, not merely one with matching numbers. For the ordinary, all-distinct case, this guarantee is free. For the degenerate, duplicate-value case, it does not exist without an id. That is the honest scope of “may become a possibility with ids” — a fix for one specific, real ambiguity, not a general precondition for UnCollapse to work at all. *[This precise scoping is the one piece of v9 directly visible in the current paper, cited by name in UnCollapse’s own section.]*

v10 — the full verb inventory settled (historical; this is the version that was drafted)

Final symbol assignment, reasoned rather than aesthetic. Proposed: $\ominus$ moves from Remove to UnCollapse; Remove takes $\backslash$ instead. Not merely a preference — $\oplus$ and $\ominus$ are a matched shape-family by standing convention (same circle, opposite sign), and that pairing belonged to Gather/Remove only while $\oplus$ meant Gather. Now that $\oplus$ means Collapse, the natural partner follows it: $\oplus/\ominus$ as Collapse/ UnCollapse restores the convention rather than breaking it. $\uplus$ (Bundle) never had a natural circled-minus partner; $\backslash$, set difference, fits it better and matches Bundle’s own description as practically a set operation. Checked clean against the source: zero instances of $\backslash\text{setminus}$ anywhere in Paper 1.

Final table:

Verb	Symbol	Produces	Status
Bundle	$\uplus$	bowl, atoms kept separate	Settled
Remove	$\setminus$	undoes Bundle	Settled (was $\ominus$)
Collapse	$\oplus$	one atom, count-preserving	Settled, this entry
UnCollapse	$\ominus$	removes one contribution	Settled, this entry
MixMerge	$\boxplus$	one atom, fused	Settled (rename only)
UnMix	$\boxminus$	—	Proven impossible, Section 7.2

Bundle’s idempotence: confirmed, already proven, nothing new. $A \uplus A = A$. Not a new claim — stated, word for word, under the retiring name: “ $\oplus$ collects: identical tokens do not double ($A \oplus A = A$)... $\{A\} \cup \{A\} = \{A\}$.” Ordinary set theory already backs this; a set never had a duplicate-counting mechanism to begin with, so $\{A_1, A_1\}$ and $\{A_1\}$ were never different objects. Confirmed by direct derivation from the “asleep on the key” case: many accidental repeats of one atom, Bundled, still return the same single bowl entry.

Collapse’s idempotence: resolved as genuinely different from Bundle’s, after real back-and-forth. $A \text{ Collapse } A \neq A$ in general. Reached by three convergent arguments, not one:

1. *Information-loss.* Collapse’s own stated purpose is that no information is lost. If $A \text{ Collapse } A = A$, the second contribution is discarded exactly the way Bundle already discards duplicates — which would make Collapse behave identically to Bundle in this one case, undercutting the reason the two needed to be separate verbs.
2. *Algebraic contradiction.* If $A \text{ Collapse } A = A$, then UnCollapsing one A back out must return the other A , giving $A \text{ UnCollapse } A = A$ — which directly contradicts $A \text{ UnCollapse } A = 0$ (the atom-level analogue of $\{A\} \ominus A = \emptyset$, independently solid). Both cannot hold unless $A = 0$.
3. *The evidence-count parallel.* This is the same shape of failure that motivated fixing Mix’s cap with the Hill form: if exact duplicates are silently discarded, two independent contributions that happen to agree exactly become indistinguishable from one contribution stated once — precisely the count-blindness that was expensive to fix once already, now reappearing in a different verb.

[**One self-correction recorded honestly.**] An early argument for this conclusion leaned on $T_w \circ T_w \neq T_w$ (repeated visits in a path are genuine, separate steps) as a direct analogy. That analogy was flagged and partly withdrawn: a path’s repeated visits are sequential, temporally separate events, while Collapse’s multiple contributions are simultaneous and co-present — not obviously the same kind of multiplicity. The three arguments above do not depend on that analogy and are treated as the actual basis for the conclusion.

Where N lives, and why nothing new needs storing. N is not a new field. Axiom 7 already defines it as the length of the bundle’s own v_i list — “ N is the atom’s info number.” If duplicates are not discarded, N is simply however many entries remain; the count was never anything other than a property of the list itself. $A \text{ Collapse } A$ gives $N = 2$: two entries, both reading A ’s own $(r, \hat{n}, \chi)$ identically. Shorthand notation for this case, proposed and worth keeping: $A^{(N)}$ for N identical copies, layered on top of the general bundle notation rather than replacing it.

[**UnCollapse’s algebra, now fully consistent.**]

$A \text{ UnCollapse } A = 0$ ($N = 1$ case, matches $\{A\} \ominus A = \emptyset$); $N \rightarrow N - 1$ per removal, in general.

$A^{(2)}$ UnCollapse $A = A^{(1)} = A$ holds correctly — specifically because $2 - 1 = 1$, not because of any general collapsing-to-bare- A rule. No remaining contradiction anywhere in the chain.

What this leaves genuinely open, not resolved here. The τ case — A Collapse B where A and B share magnitude and position but differ in sort — is not handled by χ : χ ’s confirmed job is disambiguating multiple things at one shared direction, nothing about sort, and current Axiom 7 requires every vector in one bundle to share the atom’s single τ . Handling genuinely different- τ contributions needs the per-vector τ_i extension flagged several entries back as not yet adopted. Not solved by anything in this entry. *[This is now a formally tracked open item in its own right; see Section 4 above.]*

17 Considered and found to need nothing filed

For completeness: the reference-atom and precursor-witness mechanism, the bundle-vector / uncapped V_{derived} distinction from Mix, the $\pi = (x; w_1, \dots, w_n)$ poem trajectory, and the precise definition of χ as internal phase-color (already stated in Axiom 1: “two identities may occupy the same direction without overwriting one another”) were all discussed at length and all turned out to already be fully present in Inference and Paper 1 respectively. Nothing from those threads is parked here, since there was nothing left unresolved to park. Checked again against `PA_ReWrite_V15.22.tex` specifically: all four are present, correctly stated, and use current terminology throughout; nothing here needed a symbol or status correction.